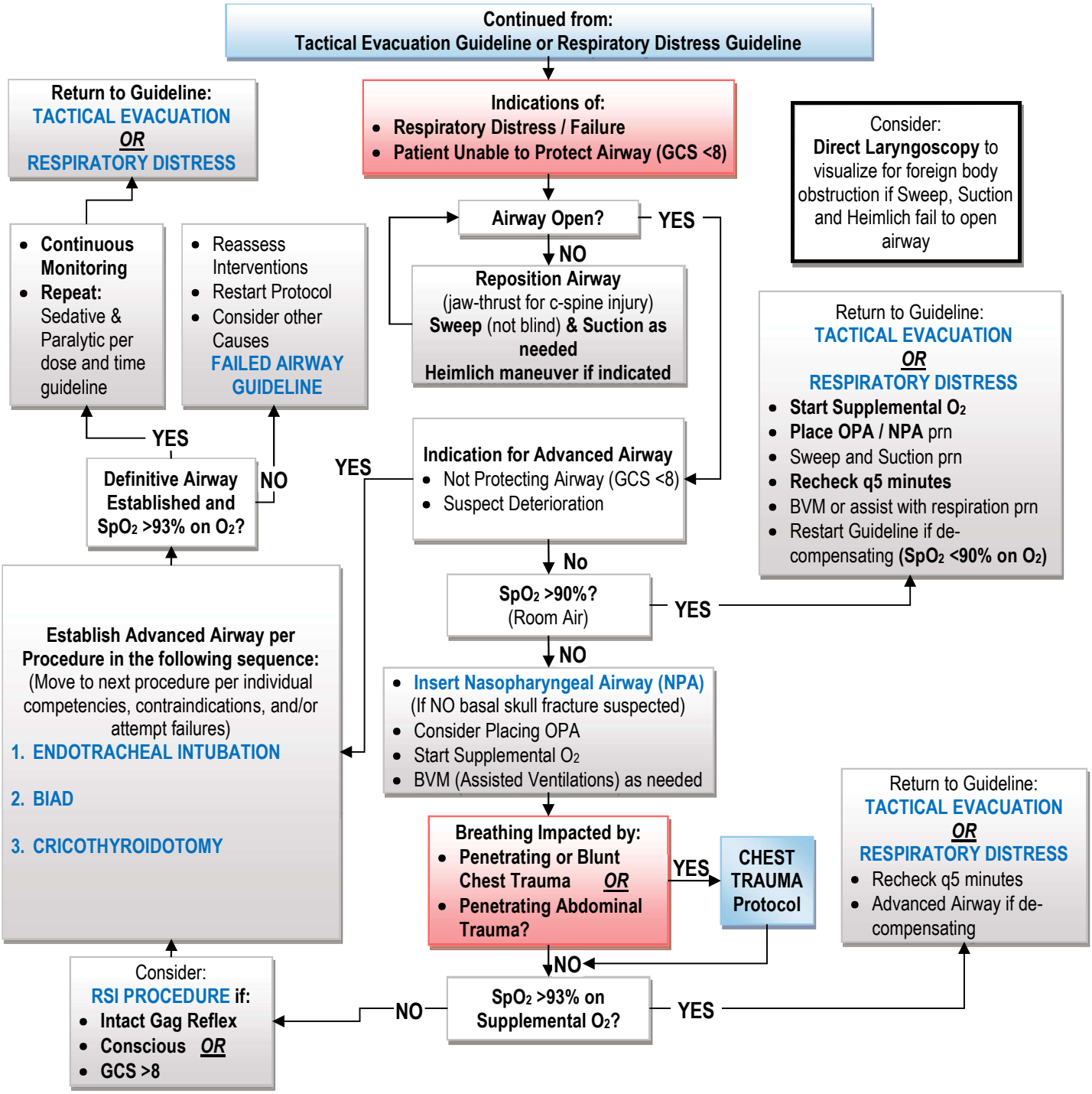
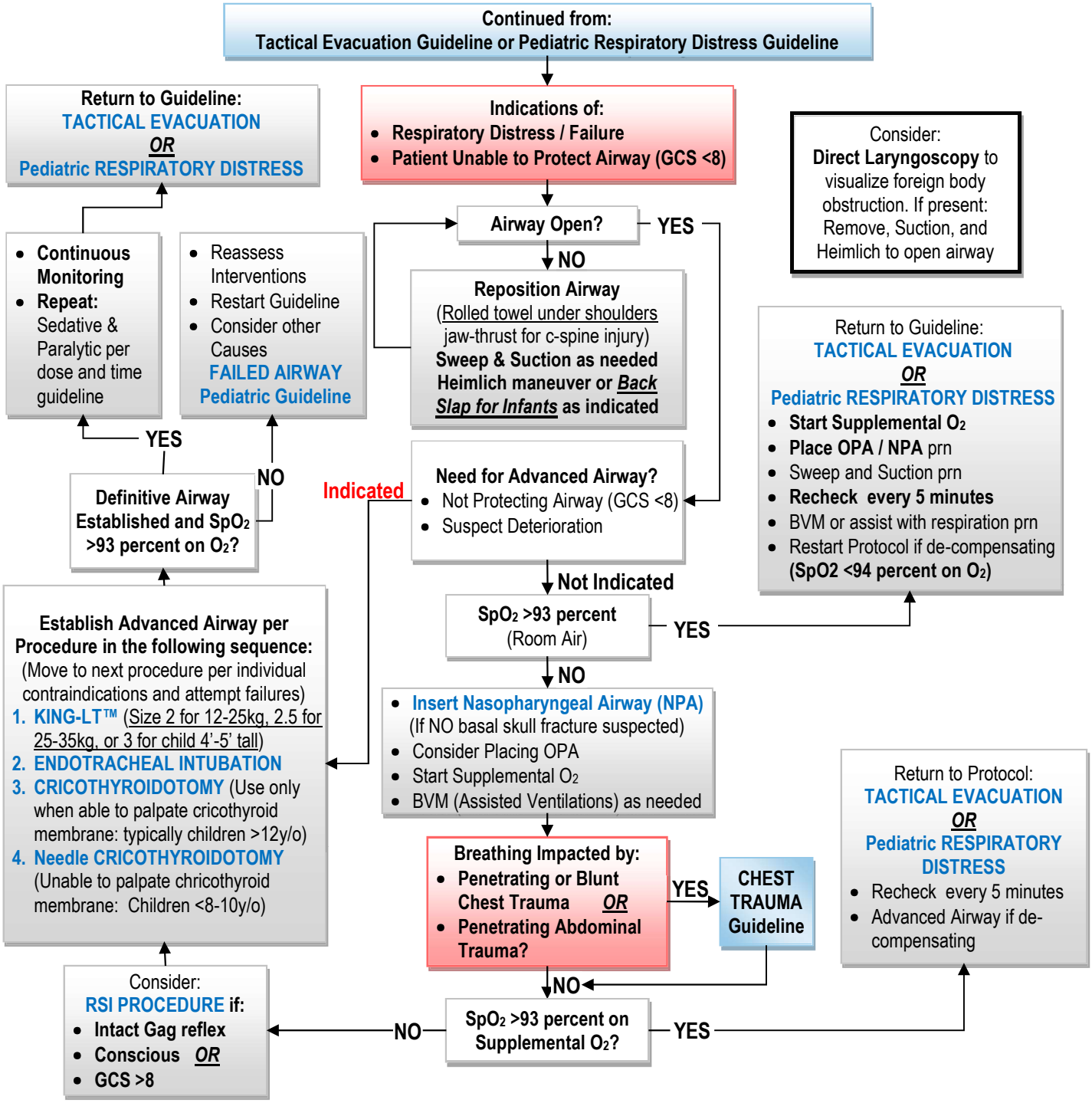
AIRWAY

Signs and Symptoms of Distress and/or Failure:

- **SpO_2 Decreasing or <90%** (Room Air) with / without supporting Signs / Symptoms of:
  - Tachypnea, Tachycardia, Fever, Cough, Wheezing, Rhonchi, Rales, Shock
- **Difficulty Breathing or Excess Work** as demonstrated by:
  - Pursing of Lips, Accessory Muscle Involvement, Cyanosis, Decreased Ability to Speak, Diaphoresis
- **Airway Obstruction** due to Trauma, Edema, Excess Secretions, Foreign Body, or Tongue
- **Apnea**
- **Decreased LOC** (GCS <8)



PEDIATRIC AIRWAY



AIRWAY Pearls

Signs and Symptoms of Respiratory Distress and/or Failure

- **SPO_2 decreasing <90%** (Room Air) with / without supporting Signs / Symptoms of:
  - Tachypnea, Tachycardia, Fever, Cough, Wheezing, Rhonchi, Rales, Shock
- **Difficulty Breathing or Excess Work** as demonstrated by:
  - PuPAIng of Lips, Accessory Muscle Involvement, Cyanosis, Decreased Ability to Speak, Diaphoresis, Tripod Breathing
- **Airway Obstruction** Due to Trauma, Edema, Excess Secretions, Foreign Body, or Tongue
- **Apnea**
- **Cyanosis,** Central and/or Peripheral: Blue/Pale Tinting and Mottling of Skin
- **Decreased LOC** (GCS <8), **Altered Responsiveness, Weak Cry**

Pearls:

- **PCO_2 is affected by respiratory rate and tidal volume (ventilation), while PO_2 is affected by PEEP and FiO_2 (oxygenation)**
- Capnography is mandatory for all intubations. Record results. Capnometer (standalone END TIDAL CO_2 detector) is an alternate if monitor capnography not available**. For capnography, normal range is 35-45 mm Hg;** adjust vent as needed.
- All intubated patients should receive nasogastric / orogastric tube (time permitting) and continuous pulse oximetry.
- **Maternal Medication:** Adverse effects can include respiratory insult to newborn.
- Pediatric is defined as anyone <12yo.
- **If RSI is impractical or provider is not credentialed to perform**, but patient requires an advanced airway with / without ventilatory support, consider:
  1. Pharmacologically-Assisted Sedation using KETAMINE followed by supraglottic airway device placement (do not attempt BIAD placement without sedation in semi-conscious patients)
  2. Surgical cricothyroidotomy using approved device. (modified 6.0 ET not ideal)
  3. Medical personnel should not actively seek to determine if gag reflex is present by touching the palate, posterior lounge, or posterior pharynx.

RSI MEDICATIONS: IV/IO Doses

Pretreatment:

Fentanyl 3mcg/kg IV

Atropine 0.02mg/kg IV Min: 0.1mg (**Infants <1yo)**

Induction Agents:	**80kg adult dose:**
Etomidate 0.3mg/kg	24mg
*Ketamine 1-2mg/kg	80-160mg
Midazolam 0.1mg/kg	8mg

Paralytics:

Vecuronium 0.08-0.15 mg/kg

*Rocuronium 1mg/kg, q25-40min

Succinylcholine 1.5mg/kg

Continued Sedation:

Fentanyl 0.5-2mcg/kg, q30-60min

Ketamine 1-2mg/kg, q10-20min

Midazolam 0.1mg/kg-NO Pain Control, q15-30m

Propofol 0.5-1.5mg/kg-NO Pain Control, q5-10m

*** =** Preferred medication for Battlefield Trauma Patients

RSI (Abbreviated: see RSI PROCEDURE as needed)

1. **Preoxygenate** (100% FiO_2 via mask or PPV as needed)
2. **Pretreat** (Premedicate) as able or mission allows (Atropine blocks reflex bradycardia in pediatric (<2y/o only) population)
3. **Induce** (Primary Sedation / Anesthesia)
4. **Paralyze** (Neuromuscular blocking agent)
5. **Wait for Fasciculation, Jaw Relaxation, Absence of Movement**
6. **Pass ET Tube or insert BIAD** (throughout attempt, ensure good O_2 saturation. If below 94% stop and provide PPV)
7. **Confirm Placement and Secure Tube**
8. **Continue Sedation and Paralytic** as needed per dosing time.

Note: Midazolam and Propofol should only be used for continued sedation when pain management is NOT a concern (i.e., Non Trauma Patient or Patient is already on adequate narcotic pain control).

Rescue Breathing Ventilation Rate Without Advanced Airway:

- **NEWBORN** = 40-60/min when performed without compressions
- **Infant / Child** = 1 breath / 3-5 seconds
- **Adult** = 1 breath / 5-6 seconds

VOCAL CORD VISUALIZATION MANEUVERS:

- **Ensure correct alignment-** External auditory meatus is aligned with sternal notch and head is in neutral to sniffing position.
- **BURP** = Backward; Upward; Rightward; Pressure on thyroid cartilage.

VENTILATOR SETTINGS:

- Mode: AC or SIMV
- Rate: Varies with age (Typical adult start rate is 12, then adjust PRN)
- Tidal Volume: 6mL/kg initially.
- I:E = 1:2-4
- PEEP: 5
- FiO_2: 90%-100% - adjust as necessary. Try to decrease FiO_2 as much as possible while keeping O_2 saturation > 93%.
- **Goal FiO_2 = 50-60% to conserve battery life and O_2, while maintaining patient SpO_2 >93%.**